



EXECUTIVE IPSEC SECURITY BRIEFING

# Executive IPsec VPN Security & Risk Posture

High-level strategic risk evaluation, cryptographic posture, and operational exposure briefing for leadership

## Section 1: Executive Summary & System Metadata

Session timestamps, endpoints evaluated, and overall audit verdict

### SECURITY HARDENED PASS

NONE · COMPLIANT

Forensic inspection validates full NIST SP 800-77 compliance. AEAD encryption, authenticated DH key exchange, and sequence counter progression verified with zero defects.

#### SESSION TIMESTAMPS

**142.08 sec**

PCAP Capture: 0.00s to 142.08s (142.08 sec)

#### ORIGIN GATEWAY

**Field Operations Client (Ubuntu Edge)**

100.119.32.83 • StrongSwan Client Daemon

#### SECURED TUNNEL

**IPsec ESP over UDP 4500 (NAT-T) · AES-256-GCM / ECP-256**

AES-256-GCM / ECP-256

#### REMOTE GATEWAY

**Enterprise Datacenter Core Gateway**

100.127.207.119 • StrongSwan Core IPsec Responder

## Section 2: Holistic Risk Scorecard

Visual 0–100 score gauge, NIST compliance status badge, and deployment readiness rating

#### COMPOSITE RISK INDEX



#### NIST SP 800-77 REV. 1 COMPLIANCE

**PASS**

NIST SP 800-77 Rev. 1

#### CRYPTOGRAPHIC RESILIENCE

**98%**

AES-GCM active, PFS missing

#### COMPLIANCE ALIGNMENT

**95%**

NIST SP 800-77 gap identified

#### DEPLOYMENT READINESS RATING

**READY FOR PRODUCTION**

Production readiness verified under federal cybersecurity guidelines.

#### THREAT EXPOSURE INDEX

**LOW**

Potential retroactive decrypt

#### CHANNEL STABILITY

**99.9%**

Zero dropped packets detected



## THREAT EXPOSURE & BUSINESS IMPACT EVALUATION

# Executive Threat Matrix

## Comprehensive register mapping discovered cryptographic and protocol vulnerabilities to direct business risk

## Section 3: Executive Threat Matrix

## Grid mapping critical vulnerabilities to business risk and operational exposure vectors

## Wiretap Eavesdropping & Retrospective Decryption

Sev: HIGH

Likelihood: Low

Adversaries recording encrypted transit can retrospectively decipher all historic corporate traffic if private keys or static PSK credentials are leaked or subpoenaed.

## ACTIVE GAP

## Credential Cracking & Identity Impersonation

Sev: MEDIUM

Likelihood: Moderate

Symmetric shared keys lack individual attribution and are vulnerable to dictionary attacks, insider theft, and rogue gateway impersonation.

**OBSERVED**

## Traffic Shape Fingerprinting & Reconnaissance

Sev: MEDIUM

Likelihood: High

Passive eavesdroppers observing packet length clustering and inter-arrival timing can accurately infer inner application types, operational cadences, and high-value data transfers.

**OBSERVED**

## Protocol Downgrade & Replay Injection

Sev: LOW

Likelihood: Low

Sequence number validation prevents packet duplication and unauthorized state re-injection. Evaluated tunnel shows 100% monotonic sequence progression.

**PROTECTED**



TRAFFIC CLASSIFICATION & STRATEGIC ROADMAP

## Traffic Overview & Strategic Remediation

Deep payload inference distribution, side-channel leakage, and prioritized resource allocation roadmap

### Section 4: High-Level Traffic Overview

Summary distribution of classified tunnel traffic and covert channel risk



### Section 5: Strategic Remediation Action Plan

Prioritized recommendations for leadership sign-off and operational resource allocation



### Executive Sign-Off & Platform Attestation



